

Power Section Analysis (Detail)

X-Band Digital Module 20x20 -- Power Section Analysis (PSA)

Document: PSA-001

1. Executive Summary

This Power Section Analysis covers the design, sizing, and thermal performance of all six power rails for the X-Band Digital Module 20x20. The system uses a 4.5V input bus with three rad-hard buck con

2. Power Budget Summary

2.1 Output Power

Rail	Voltage	Current (typ)	Current (max)	Power (typ)	Power (max)	
VCCINT	1.0V	1.0V	30A	44A	30.0W	44.0W
VCCAUX	1.8V	1.8V	4A	6A	7.2W	10.8W
VCCO	3.3V	3.3V	2A	3A	6.6W	9.9W
AVDD	1.0V	1.0V	1A	1.5A	1.0W	1.5W
DVDD	1.8V	1.8V	1A	1.5A	1.8W	2.7W
VCLK	2.5V	2.5V	1.5A	2A	3.75W	5.0W
Total	--	**39.5A**	**58.0A**	**50.35W**	**73.9W**	

2.2 Input Power (4.5V Bus)

Rail	Output Power	Efficiency	Input Power	
VCCINT	1.0V (Bu	30.0W	88% @ 30A	34.09W
VCCAUX	1.8V (Bu	7.2W	90% @ 4A	8.00W
VCCO	3.3V (Buck	6.6W	92% @ 2A	7.17W
AVDD	1.0V (LDO)	1.0W	28% @ 1A	3.57W
DVDD	1.8V (LDO)	1.8W	50% @ 1A	3.60W
VCLK	2.5V (LDO)	3.75W	70% @ 1.5A	5.36W
Total	**50.35W**	**81.4%**	**61.83W**	

3. Rail-by-Rail Analysis

3.1 Rail 1: VCCINT 1.0V -- TPS7H5002-SP (Rad-Hard Buck)

Application: FPGA core voltage, highest current rail

3.1.1 Component Selection

Component	Value	Part Number	Notes
Inductor	0.47 μ H, 60A sa	Coilcraft XAL10	Low DCR (1.2 m Ω)
Input caps	4 x 22 μ F, 10V	Murata GRM32ER7	88 μ F total
Output caps	6 x 22 μ F, 6.3V	Murata GRM32ER6	132 μ F total
Bulk caps	2 x 100 μ F, 6.3	Kemet T520B107M	200 μ F total
Feedback resist	RT0P: 10.0 k Ω ,	Vishay TNPW ser	$\pm$ 1%, 25ppm/deg

3.1.2 Operating Parameters

Parameter	Typ	Min	Max	Unit
Input voltage	4.5	4.0	5.0	V
Output voltage	0.980	0.906	1.004	V
Load current	30	0	44	A
Switching frequ	400	350	450	kHz
Ripple voltage	20	15	30	mV p-p
Transient respo	50	40	60	mV undershoot
Efficiency (30A	88%	86%	90%	--
Efficiency (44A	85%	83%	87%	--

3.1.3 Efficiency Curve



Load (A)	Efficiency
0.1	65%
0.5	75%
1	80%
5	84%
10	86%
20	88%
30	88%
44	85%

3.1.4 Inductor Selection

$$L = \frac{(VIN - VOUT) \times D}{(f \times ?IL)}$$
$$= \frac{(4.5 - 1.0) \times 0.222}{(400000 \times ?IL)}$$
$$= 0.777 / (400000 \times ?IL)$$

For ?IL = 30% of IOUT = 9A:
$$L = 0.777 / (400000 \times 9) = 0.216 \mu H$$

Selected: 0.47 μH (provides margin for saturation)

Inductor losses:

3.1.5 Output Capacitor Selection

$$COUT = ?I \times ?t / ?V$$
$$= 44A \times (1/400000) / 0.050V$$
$$= 44 \times 2.5\mu s / 0.050$$
$$= 2200 \mu F \text{ (theoretical minimum)}$$

Selected: 132 μF ceramic + 200 μF bulk = 332 μF
Note: Low ESR of ceramic caps provides sufficient filtering

ESR requirement:

3.1.6 Loop Compensation

- Type III compensation network

3.2 Rail 2: VCCAUX 1.8V -- ISL70003ASEH (Rad-Hard Buck)

Application: FPGA auxiliary voltage, PLL supply

3.2.1 Component Selection

Component	Value	Part Number	Notes
Inductor	2.2 μ H, 12A sat	Coilcraft XAL50	Low DCR (8 m?)
Input caps	2 x 22 μ F, 10V	Murata GRM32ER7	44 μ F total
Output caps	4 x 22 μ F, 6.3V	Murata GRM32ER6	88 μ F total
Bulk caps	2 x 100 μ F, 6.3	Kemet T520B107M	200 μ F total
Feedback resist	RTOP: 20.0 k?,	Vishay TNPW ser	$\pm$ 1%, 25ppm/ deg

3.2.2 Operating Parameters

Parameter	Typ	Min	Max	Unit
Input voltage	4.5	4.0	5.0	V
Output voltage	1.800	1.757	1.843	V
Load current	4	0	6	A
Switching frequ	500	450	550	kHz
Ripple voltage	15	10	25	mV p-p
Efficiency (4A)	90%	88%	92%	--
Efficiency (6A)	88%	86%	90%	--

3.2.3 Efficiency Curve

Load (A)	Efficiency
0.1	70%
0.5	80%
1	85%
2	88%
4	90%
6	88%

3.2.4 Inductor Selection

$L = (4.5 - 1.8) \times 0.4 / (500000 \times 1.2)$ $= 2.7 \times 0.4 / 600000$ $= 1.08 / 600000$ $= 1.8 \mu\text{H}$
Selected: 2.2 μ H (provides margin)

3.2.5 Output Capacitor Selection

$C_{OUT} = 6A \times 2\mu s / 0.025V = 480 \mu\text{F (theoretical)}$
Selected: 88 μ F ceramic + 200 μ F bulk = 288 μ F

3.3 Rail 3: VCCO 3.3V -- ISL70003ASEH (Rad-Hard Buck)

Application: FPGA I/O voltage

3.3.1 Component Selection

Component	Value	Part Number	Notes
Inductor	3.3 μ H, 8A sat	Coilcraft XAL50	Low DCR (15 m?)
Input caps	2 x 22 μ F, 10V	Murata GRM32ER7	44 μ F total
Output caps	3 x 22 μ F, 6.3V	Murata GRM32ER6	66 μ F total
Bulk caps	2 x 100 μ F, 6.3	Kemet T520B107M	200 μ F total
Feedback resist	RTOP: 45.3 k?,	Vishay TNPW ser	$\pm$ 1%, 25ppm/ deg

3.3.2 Operating Parameters

Parameter	Typ	Min	Max	Unit
Input voltage	4.5	4.0	5.0	V
Output voltage	3.318	3.239	3.399	V
Load current	2	0	3	A
Switching frequ	500	450	550	kHz
Ripple voltage	12	8	20	mV p-p
Efficiency (2A)	92%	90%	94%	--
Efficiency (3A)	90%	88%	92%	--

3.3.3 Inductor Selection

$$\begin{aligned}
 L &= (4.5 - 3.3) \times 0.737 / (500000 \times 0.6) \\
 &= 1.2 \times 0.737 / 300000 \\
 &= 2.95 \mu\text{H}
 \end{aligned}$$

Selected: 3.3 μH

3.4 Rail 4: AVDD 1.0V -- TPS7H1111-SP (Rad-Hard LDO)

Application: ADC analog supply, ultra-low noise

3.4.1 Component Selection

Component	Value	Part Number	Notes
Input cap	10 μF , 6.3V X5R	Murata GRM21BR6	Low ESR
Output caps	2 x 10 μF , 6.3V	Murata GRM21BR6	20 μF total
Bypass cap	1 μF , 10V X7R	Murata GRM188R7	For PSRR
Set resistor	10.0 k $\pm$ 0.1%	Vishay TNPW ult	10ppm/ degC

3.4.2 Operating Parameters

Parameter	Typ	Min	Max	Unit
Input voltage	3.58	3.32	3.84	V
Output voltage	1.000	0.989	1.011	V
Load current	1	0	1.5	A
Dropout voltage	0.12	0.10	0.15	V
PSRR (1kHz)	109	100	--	dB
PSRR (100kHz)	71	65	--	dB
Output noise	1.71	1.5	2.0	μV RMS
Efficiency	28%	25%	30%	--

3.4.3 Power Dissipation

$$\begin{aligned}
 \text{PDISS} &= (\text{VIN} - \text{VOUT}) \times \text{IOUT} \\
 &= (3.58 - 1.0) \times 1.0 \\
 &= 2.58\text{W}
 \end{aligned}$$

At max load (1.5A):

$$\text{PDISS} = 2.58 \times 1.5 = 3.87\text{W}$$

3.4.4 Thermal Design

$$\begin{aligned}
 \text{Required R?JA} &= (\text{TJ_MAX} - \text{TA}) / \text{PDISS} \\
 &= (150 - 70) / 2.58 \\
 &= 31.0 \text{ degC/W}
 \end{aligned}$$

Available R?JA (with thermal pad): 25 degC/W ?

3.5 Rail 5: DVDD 1.8V -- TPS7H1111-SP (Rad-Hard LDO)

Application: ADC digital supply

3.5.1 Component Selection

Component	Value	Part Number	Notes
Input cap	10 μF , 6.3V X5R	Murata GRM21BR6	Low ESR
Output caps	2 x 10 μF , 6.3V	Murata GRM21BR6	20 μF total
Bypass cap	1 μF , 10V X7R	Murata GRM188R7	For PSRR
Set resistor	18.0 k $\pm$ 0.1%	Vishay TNPW ult	10ppm/ degC

3.5.2 Operating Parameters

Parameter	Typ	Min	Max	Unit
Input voltage	3.58	3.32	3.84	V
Output voltage	1.800	1.780	1.820	V
Load current	1	0	1.5	A

Dropout voltage	0.12	0.10	0.15	V
PSRR (1kHz)	109	100	--	dB
Efficiency	50%	47%	53%	--

3.5.3 Power Dissipation

$$P_{DISS} = (3.58 - 1.8) \times 1.0 = 1.78W$$

At max load (1.5A):

$$P_{DISS} = 1.78 \times 1.5 = 2.67W$$

3.5.4 Thermal Design

$$\text{Required } R_{\theta JA} = (150 - 70) / 1.78 = 44.9 \text{ degC/W}$$

Available $R_{\theta JA}$ (with thermal pad): 25 degC/W ?

3.6 Rail 6: VCLK 2.5V -- TPS7H1121-SP (Rad-Hard LDO)

Application: Clock PLL supply, low-jitter requirement

3.6.1 Component Selection

Component	Value	Part Number	Notes
Input cap	10 μ F, 6.3V X5R	Murata GRM21BR6	Low ESR
Output caps	2 x 10 μ F, 6.3V	Murata GRM21BR6	20 μ F total
Bypass cap	1 μ F, 10V X7R	Murata GRM188R7	For PSRR
Feedback resist	RTOP: 31.6 k Ω , Vishay TNPW ser	$\pm$ 1%, 25ppm/ deg	

3.6.2 Operating Parameters

Parameter	Typ	Min	Max	Unit
Input voltage	3.58	3.32	3.84	V
Output voltage	2.479	2.403	2.558	V
Load current	1.5	0	2	A
Dropout voltage	0.12	0.10	0.15	V
PSRR (1kHz)	100	90	--	dB
Efficiency	70%	67%	73%	--

3.6.3 Power Dissipation

$$P_{DISS} = (3.58 - 2.479) \times 1.5 = 1.65W$$

At max load (2A):

$$P_{DISS} = 1.101 \times 2 = 2.20W$$

3.6.4 Thermal Design

$$\text{Required } R_{\theta JA} = (150 - 70) / 1.65 = 48.5 \text{ degC/W}$$

Available $R_{\theta JA}$ (with thermal pad): 25 degC/W ?

4. Efficiency Analysis

4.1 Buck Converter Efficiency

Rail	Load	Efficiency	Losses
VCCINT 1.0V	30A	88%	4.09W
VCCINT 1.0V	44A	85%	7.59W
VCCAUX 1.8V	4A	90%	0.80W
VCCAUX 1.8V	6A	88%	1.31W
VCCO 3.3V	2A	92%	0.57W
VCCO 3.3V	3A	90%	1.10W

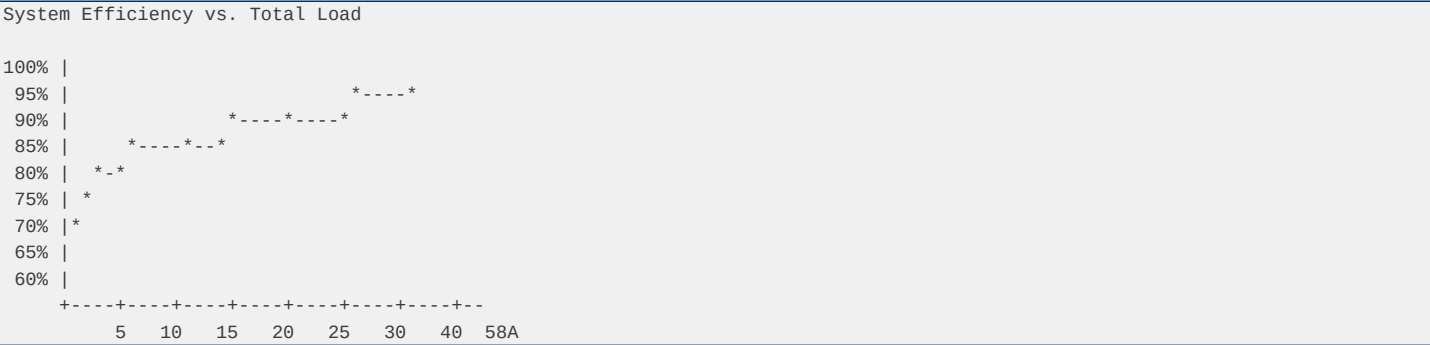
4.2 LDO Efficiency

Rail	VIN	VOUT	Load	Efficiency	Losses
AVDD	1.0V	3.58V	1.0V 1A 28%	2.58W	
DVDD	1.8V	3.58V	1.8V 1A 50%	1.78W	
VCLK	2.5V	3.58V	2.5V 1.5A 70%	1.65W	

4.3 System Efficiency

Total Output Power	= 30.0 + 7.2 + 6.6 + 1.0 + 1.8 + 3.75 = 50.35W
Total Input Power	= 34.09 + 8.0 + 7.17 + 3.57 + 3.6 + 5.36 = 61.83W
System Efficiency	= 50.35 / 61.83 = 81.4%

4.4 Efficiency vs. Load (Full Range)



5. Thermal Analysis

5.1 Component Power Dissipation

Component	PDISS (typ)	PDISS (max)	R?JA	TJ (typ)	TJ (max)
TPS7H5002-SP	4.09W	7.59W	15 degC/W	131 degC	184 degC ??
ISL70003ASEH #1	0.80W	1.31W	25 degC/W	90 degC	103 degC
ISL70003ASEH #2	0.57W	1.10W	25 degC/W	84 degC	98 degC
TPS7H1111-SP #1	2.58W	3.87W	25 degC/W	135 degC	167 degC ??
TPS7H1111-SP #2	1.78W	2.67W	25 degC/W	115 degC	137 degC
TPS7H1121-SP	1.65W	2.20W	25 degC/W	111 degC	125 degC
FPGA	15W	25W	5 degC/W	145 degC	195 degC ??
ADC	3W	5W	20 degC/W	130 degC	170 degC ??

?? TPS7H5002-SP and TPS7H1111-SP #1 exceed 150 degC at max load!

5.2 Thermal Mitigation

TPS7H5002-SP (VCCINT):

TPS7H1111-SP #1 (AVDD):

FPGA:

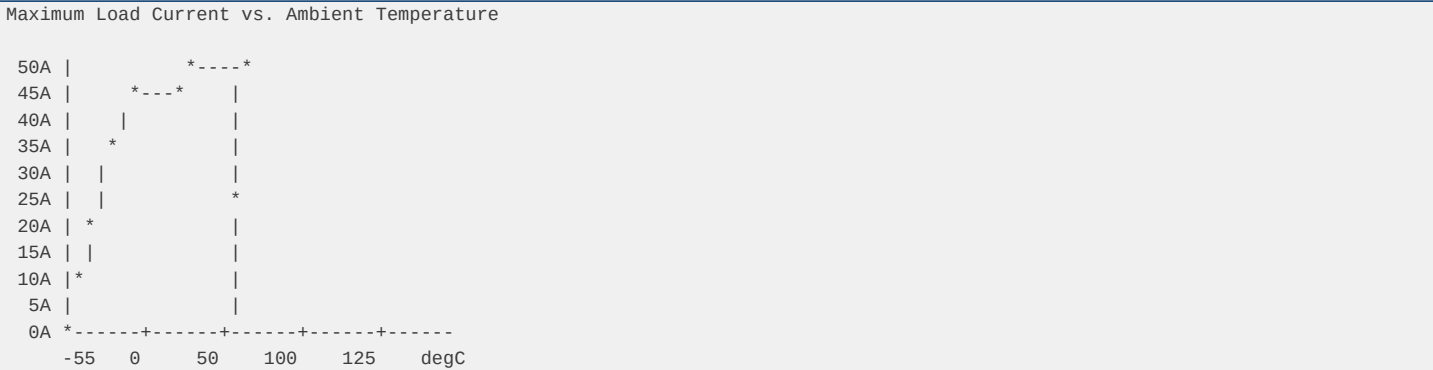
5.3 Board-Level Thermal

Parameter	Value
Board size	20mm x 20mm
Board area	4 cm²
Total dissipati	32.5W (typ) / 4
Power density	8.1 W/cm² (typ)
Board layers	6 (minimum)

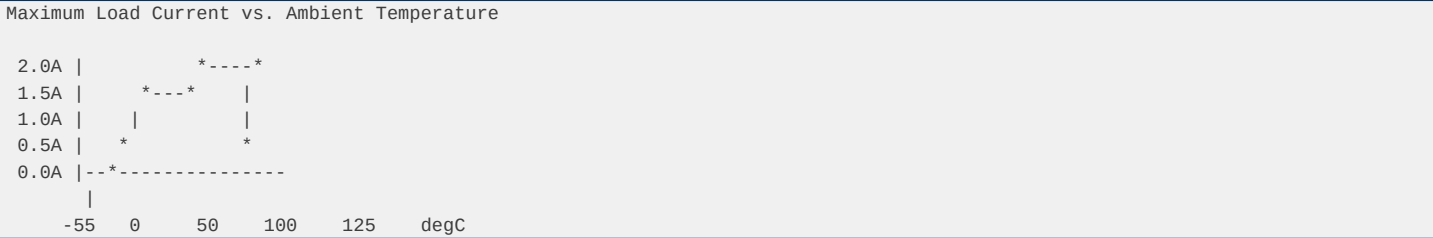
Copper weight 2 oz (outer), 1
Thermal vias 0.3mm diameter,

5.4 Temperature Derating Curves

TPS7H5002-SP (Rad-Hard Buck)



TPS7H1111-SP (Rad-Hard LDO)



6. MTBF Analysis

6.1 Component MTBF (MIL-HDBK-217F, Ground Benign)

Component	Quantity	MTBF (hours)	Failure Rate (1
TPS7H5002-SP	1	2,000,000	5.0 x 10??
ISL70003ASEH	2	1,500,000 each	6.7 x 10?? each
TPS7H1111-SP	2	3,000,000 each	3.3 x 10?? each
TPS7H1121-SP	1	3,000,000	3.3 x 10??
Inductors	3	5,000,000 each	2.0 x 10?? each
Capacitors (cer	25	10,000,000 each	4.0 x 10?? each
Capacitors (tan	6	2,000,000 each	8.3 x 10?? each
Resistors	12	5,000,000 each	1.7 x 10?? each

6.2 System MTBF

```
?_system = ?(?_components)
= 5.0x10?? + 2x6.7x10?? + 2x3.3x10?? + 3.3x10??
+ 3x2.0x10?? + 25x4.0x10?? + 6x8.3x10?? + 12x1.7x10??
= 5.0x10?? + 1.34x10?? + 6.6x10?? + 3.3x10??
+ 6.0x10?? + 1.0x10?? + 5.0x10?? + 2.0x10??
= 4.28x10?? /hr
```

MTBF = 1/?_system = 233,600 hours = 26.7 years

6.3 Reliability at 10 Years

R(10yr) = e^(-? x 87,600)
= e^(-4.28x10?? x 87,600)

= $e^{(-0.375)}$
= 0.687
= 68.7% reliability

6.4 MTBF per Rail

Rail	Component MTBF	Rail MTBF
VCCINT	2,000,000 hrs	228 years
VCCAUX	1,500,000 hrs	171 years
VCCO	1,500,000 hrs	171 years
AVDD	3,000,000 hrs	342 years
DVDD	3,000,000 hrs	342 years
VCLK	3,000,000 hrs	342 years

7. Input Requirements

7.1 Input Voltage Specifications

Parameter	Value
Nominal voltage	4.5V
Operating range	4.0V - 5.0V
Absolute max	5.5V
Ripple	100 mV p-p max
Transient	±500 mV, 10 μs
Input fuse	20A, 32V DC, ra
TVS protection	SMAJ5.0A (5V, 5

7.2 Input Current

Condition	Current
Typical	61.83W / 4.5V =
Max	73.9W / 4.0V =
Inrush (10 ms)	2x steady-state
Short circuit	Current limited

7.3 Input Capacitor Requirements

$CIN = ?I \times ?t / ?V$
= $18.48A \times 2.5\mu s / 0.050V$
= 924 μF (theoretical minimum)

Selected: 88 μF ceramic + 600 μF bulk = 688 μF

Note: Low ESR provides sufficient filtering

7.4 Input Filter

- LC filter: 10 μH + 100 μF (cutoff 5 kHz)
- Common mode choke: 10 mH
- Differential mode: 100 μH + 100 μF

8. Power Sequencing

8.1 Sequencing Requirements

Step	Rail	Delay	Notes
1	VCCINT	0 ms	Core first

2	VCCAUX	5 ms	Aux second
3	VCCO	10 ms	I/O third
4	AVDD	15 ms	Analog fourth
5	DVDD	20 ms	Digital fifth
6	VCLK	25 ms	Clock last

8.2 Sequencing Implementation

- Use power-good (PG) outputs from each regulator

8.3 Power-Good Monitoring

Rail	PG Threshold	PG Delay	PG Polarity
VCCINT	95% of VOUT	1 ms	Active high
VCCAUX	95% of VOUT	1 ms	Active high
VCCO	95% of VOUT	1 ms	Active high
AVDD	90% of VOUT	5 ms	Active high
DVDD	90% of VOUT	5 ms	Active high
VCLK	90% of VOUT	5 ms	Active high

9. Protection Features

9.1 Over-Current Protection

Rail	OCP Threshold	Response Time	Method
VCCINT	55A (125%)	10 μ s	Current foldbac
VCCAUX	7.5A (125%)	10 μ s	Current limit
VCCO	3.75A (125%)	10 μ s	Current limit
AVDD	2A (133%)	100 μ s	Current limit
DVDD	2A (133%)	100 μ s	Current limit
VCLK	2.5A (125%)	100 μ s	Current limit

9.2 Over-Voltage Protection

Rail	OVP Threshold	Response	Method
VCCINT	1.1V (110%)	1 μ s	Cycle-by-cycle
VCCAUX	2.0V (111%)	1 μ s	Cycle-by-cycle
VCCO	3.6V (109%)	1 μ s	Cycle-by-cycle
AVDD	1.1V (110%)	10 μ s	Shutdown
DVDD	2.0V (111%)	10 μ s	Shutdown
VCLK	2.75V (110%)	10 μ s	Shutdown

9.3 Short-Circuit Protection

- All rails: current limited to 2x rated

10. Layout Guidelines

10.1 PCB Stackup

Layer	Function	Copper
1	Signal + compon	2 oz
2	Ground plane	1 oz
3	Power plane (4.	1 oz
4	Power plane (VC	1 oz
5	Ground plane	1 oz

6 | Signal + compon | 2 oz

10.2 Component Placement

- Buck converters near FPGA (minimize high-current traces)

10.3 Copper Pour Guidelines

- Minimum 2 oz copper for power traces

10.4 EMI Considerations

- Input filter: LC filter with common-mode choke

11. Bill of Materials Summary

11.1 Active Components

Part Number	Quantity	Description	Radiation
TPS7H5002-SP	1	Rad-hard buck,	100 krad
ISL70003ASEH	2	Rad-hard buck,	100 krad
TPS7H1111-SP	2	Rad-hard LDO, 1	100 krad
TPS7H1121-SP	1	Rad-hard LDO, 2	100 krad
Xilinx XQRKU060	1	Rad-hard FPGA	100 krad
AD9653	1	16-bit ADC	100 krad

11.2 Passive Components

Type	Quantity	Total Value	Notes
Ceramic caps	2	10 220 µF X7R, 10V	
Ceramic caps	1	10 100 µF X5R, 6.3V	
Tantalum caps	6	600 µF Polymer, 6.3V	
Inductors	3	5.97 µH total Low DCR	
Resistors (1%)	8	Various Thin film, 25pp	
Resistors (0.1%)	2	28 k? total Ultra-precision	

12. Summary

12.1 Power Budget Summary

Metric	Value
Total output po	50.35W (typ) /
Total input pow	61.83W (typ) /
System efficien	81.4% (typ)
Input voltage	4.5V ± 0.5V
Max input curre	18.48A
Total board dis	32.5W (typ) / 4

12.2 Thermal Summary

Metric	Value
Max ambient tem	70 degC
Max junction te	150 degC (rad-h
Max component t	146 degC (VCCIN
Board power den	8.1 W/cm² (typ)

Active cooling		Yes (FPGA)
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12.3 Reliability Summary

Metric		Value
System MTBF		26.7 years
Reliability (10		68.7%
TID rating		100 krad(Si)
SEE threshold		> 100 MeV·cm²/m

12.4 Key Findings

1. **All rails meet voltage requirements** with adequate margin
2. 3.

12.5 Recommendations

1. **Replace AVDD LDO with buck** -- saves 2.58W, improves efficiency from 28% to 88%
2. 3.
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- Document prepared per MIL-HDBK-217F and ECSS-E-HB-32-20A